QuantRule — Architecture Characteristics

QuantRule

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Eight characteristics matter. Each is tied to a kata requirement and has a level that can be measured. The rest of the course list is deliberately not optimised.

Characteristic	Kata	Why it matters	Level (measurable)	Measured by
Testability	R2, R3, R5	every published number is a claim someone defends	every reading and event has a causality, a determinism, and a golden test; fast suite <= 5 min	`cargo test`, gate
Modularity	R4, R6, R7	six services only pay off if boundaries hold	6 services, 0 cycles, gateway imports 0 engine symbols	boundary test
Maintainability	R3, R7	the rule changes weekly, the platform must not	a rule change touches <= 1 crate and <= 3 modules; no file over 1,000 lines	structural check
Deployability	R4, R6	a reviewer must reproduce in one command	clean clone to healthy <= 10 min; 1 published port; desktop is 1 asset	`make up`
Fault tolerance	R4, R5	a run must survive a crash without corrupting evidence	0 succeeded runs with a missing artifact; resubmission gives identical bytes	crash test
Responsiveness	R4, R7	marking is interactive, a year of candles is not	submit <= 200 ms; chart window <= 200 ms; 1 year, 8 timeframes <= 60 s on 8 cores	benchmark
Scalability	R4	one machine must borrow a second	model stage 1 to 4 workers: identical digest, >= 2.5x throughput	scale test
Partitioning type	R1-R7	the course asks for the choice explicitly	domain-driven: 6 services by capability, no layer split	service catalog

Not optimised: elasticity (no burst traffic, one operator, batch work), localization (one language, one user), and accessibility (one expert user; keyboard support is kept, but it is not a driver).